

Setup

Vocabulary: tokens: I , You , love , hate , cats , dogs .

We generate **3 tokens** (pick token1, token2, token3) after the start token.

The model gives the following conditional probabilities:

Start (token1)

- $P(I) = 0.55$
- $P(You) = 0.45$

If token1 = I (token2 probabilities)

- $P(\text{love} \mid I) = 0.6$
- $P(\text{hate} \mid I) = 0.4$

If token1 = You (token2 probabilities)

- $P(\text{love} \mid You) = 0.2$
- $P(\text{hate} \mid You) = 0.8$

If token2 = love (token3 probabilities)

- $P(\text{cats} \mid \text{love}) = 0.5$
- $P(\text{dogs} \mid \text{love}) = 0.5$

If token2 = hate (token3 probabilities)

- $P(\text{cats} \mid \text{hate}) = 0.1$
- $P(\text{dogs} \mid \text{hate}) = 0.9$

Greedy search (beam = 1)

At each step pick the most probable next token.

1. **Step 1 (token1):** pick I because $0.55 > 0.45$.
2. **Step 2 (token2) given I :** pick love because $0.6 > 0.4$.
3. **Step 3 (token3) given love :** tie between cats and dogs (both 0.5). Assume tie $\rightarrow$ pick cats .

Final greedy sequence: I love cats .

Probability = $0.55 \times 0.6 \times 0.5$.

Calculate:

- $0.55 \times 0.6 = 0.33$.
- $0.33 \times 0.5 = 0.165$.

So greedy gives I love cats with probability **0.165**.

Beam search (beam width = 2)

Keep the top 2 partial sequences at each step.

After Step 1 (keep top 2 token1):

- I (0.55)
- You (0.45)

Expand to token2 (all children of those two):

- From I :
 - I love : $0.55 \times 0.6 = 0.33$
 - I hate : $0.55 \times 0.4 = 0.22$
- From You :
 - You love : $0.45 \times 0.2 = 0.09$
 - You hate : $0.45 \times 0.8 = 0.36$

Keep top 2 partial sequences (by their joint scores):

- You hate — 0.36
 - I love — 0.33
- (We drop I hate 0.22 and You love 0.09.)

Expand those two to token3:

- Expand You hate (0.36):
 - You hate cats : $0.36 \times 0.1 = 0.036$
 - You hate dogs : $0.36 \times 0.9 = 0.324$
- Expand I love (0.33):
 - I love cats : $0.33 \times 0.5 = 0.165$
 - I love dogs : $0.33 \times 0.5 = 0.165$

Now compare final sequences:

- You hate dogs : **0.324**
- I love cats : 0.165
- I love dogs : 0.165
- You hate cats : 0.036

Beam's top sequence is You hate dogs with probability **0.324**.

Conclusion — where they differ

- Greedy (beam=1) picks I love cats with probability 0.165.
- Beam (B=2) finds You hate dogs with probability 0.324, which is higher.